

ABHAY P

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PROFESSIONAL SUMMARY

Junior AI and Computer Vision Engineer with hands-on industry experience in real-time video analytics, person re-identification, and IoT-based time-series forecasting using Python, OpenCV, and NVIDIA DeepStream. Published researcher in EEG signal classification with a strong foundation in building end-to-end AI pipelines. Eager to contribute to Python-centric engineering teams focused on AI, backend systems, or computer vision.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, C++, HTML, CSS
- **Web / Backend:** FastAPI, REST APIs, React.js, .NET
- **AI & Computer Vision:** OpenCV, NVIDIA DeepStream, Visual Re-ID, Cosine Similarity
- **Machine Learning:** iTransformer, Random Forest, K-Means, Scikit-learn, NumPy, Pandas
- **Databases:** PostgreSQL, MySQL
- **IoT & Edge:** ThingsBoard, Cumulocity, MQTT
- **DevOps & Tools:** Docker, Git, GitHub, Bitbucket, Linux, Postman

EXPERIENCE

Software Engineer

Inthings Technologies Pvt Ltd

Perinthalmanna

Sept 2025 – Present

- Developed real-time people-detection and tracking pipelines using Python, OpenCV, and NVIDIA DeepStream, enabling simultaneous multi-camera video analytics across 3+ camera streams in production environments.
- Built a multivariate time-series forecasting system using the iTransformer model on 36,000+ high-frequency IoT sensor data points across 5+ devices, generating 30-day predictions for temperature and humidity with high accuracy.
- Designed and deployed a person re-identification module using cosine similarity-based embedding matching, enabling consistent identity tracking across 4+ non-overlapping camera feeds with improved cross-camera accuracy.
- Maintained codebase quality through Git/GitHub/Bitbucket version control, code reviews, and collaborative debugging, reducing integration bugs across an agile development team.

PROJECTS

Fleet Management Platform (React + .NET)

Mar 2026 - Present

- Built a full-stack fleet management SaaS platform with a React (Vite) SPA and an ASP.NET Core 8 REST API (12 controllers), enabling real-time monitoring of IoT devices across a multi-tenant partner-organization hierarchy.
- Implemented role-based access control (RBAC) with 6 granular roles using Azure CIAM, MSAL, and JWT Bearer authentication, with claims enrichment injected at token issuance via a custom Azure extension endpoint.
- Integrated Cumulocity IoT for live device telemetry, GPS location tracking, alarm management, and device onboarding, with PostgreSQL (EF Core) as the persistent data store and Azure Key Vault for secrets management.

Person Re-Identification System (OpenCV + Embedding Model)

Dec 2025 - Feb 2026

- Engineered a cross-camera person re-identification system using bounding-box detection, deep feature extraction, and cosine similarity scoring to match individuals across multiple non-overlapping camera feeds.

- Integrated an embedding generation pipeline to create identity descriptors robust to variations in lighting, pose, and camera angle, enabling reliable real-world deployment.

Time-Series Forecasting (iTransformer Model)

Sept 2025 – Nov 2025

- Designed a multivariate forecasting pipeline using the iTransformer model, preprocessing 36,000+ high-frequency IoT data points (5-minute intervals) with engineered time features.
- Generated 30-day long-horizon predictions for temperature and humidity across multiple IoT devices, validating results against unseen test data with high accuracy.

Automated Exam Paper Evaluation [Main Project]

Aug 2024 – Apr 2025

- Developed an end-to-end AI system for automated handwritten exam evaluation using DTrOCR for handwriting recognition and GPT-based models for context-aware semantic answer grading.
- Integrated NLP-driven feedback generation to produce detailed grading reports, significantly reducing manual evaluation effort for subjective answer sheets.

Classification of EEG Signals [Mini Project]

Jan 2024 – May 2024

- Built a machine learning pipeline in Python to classify seizure stages from raw EEG signals using Random Forest and K-Means clustering with engineered signal features.
- Preprocessed and extracted discriminative features from EEG data, achieving results published at the CEMC2024 international conference.

EDUCATION

Sreepathy Institute of Management and Technology (SIMAT)

Oct 2021 – May 2025

Bachelor of Technology in Computer Science and Engineering

- GPA: 7.0 / 10

KHSS Thottara

Jun 2019 – Mar 2021

Higher Secondary Education – Computer Science Stream

- Percentage: 94%

PUBLICATIONS

Classification of EEG Signals

Jan 2025

- Published in the proceedings of *CEMC2024*, organized by IDES and Grenze Scientific Society.
- Authored and presented research on EEG signal processing and machine learning applications in biomedical engineering.